

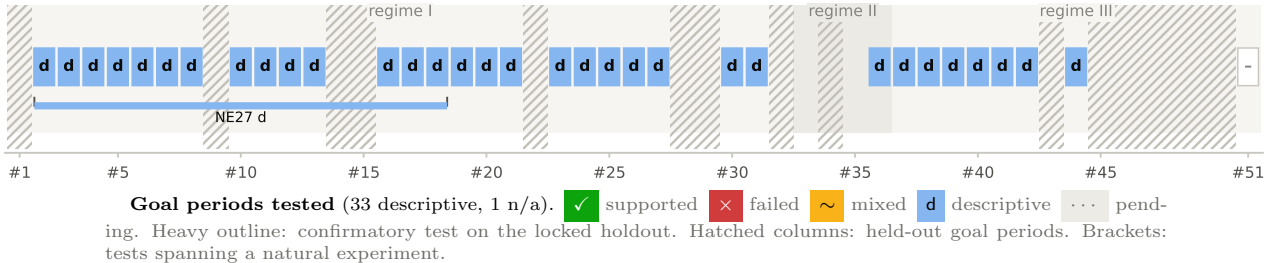
H106: A finite magnet's magnetization wanders as 1/N: the slow mode's decay time should grow with village size

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 0.5/5

Bottom line. The $1/N$ finite-magnet signature of H81's slow mode is not identifiable in one village (power 0.00, oracle SD 0.45; real $\alpha_k +0.40 / +1.22$ with CIs of ± 2), so this test does not weigh on the outside-drift rival; post hoc, the mode is confined to the $N < 8$ era.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

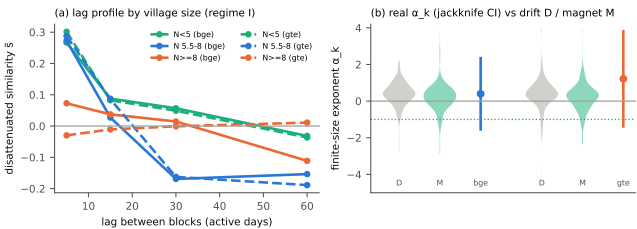
The test asks whether H81's regime-I culture slow mode has the inertia of a finite ordered magnet, whose direction wanders at a rate $\propto 1/N$, or the rate of an outside drift, which does not depend on N . Culture vectors come from random 4-agent subsets in each goal×week block (24 goals, N 4–12), with split-half disattenuation. A synthetic on the real panel, run before any real statistic, shows that the slope cannot be measured: power 0.00 against $\alpha_k = -1$, and a noiseless oracle still has SD 0.45. The real slopes are $\alpha_k = +0.40 [-1.62, 2.41]$ (bge) and $+1.22 [-1.45, 3.88]$ (gte). The NE27 jump (N 4→7) and the #51 joins are unpowered. Post hoc, the slow mode is strong before mid-November 2025 and weak or gone after it.

PHYSICAL MODEL

Vector spins (model 11) with finite-size scaling (model 14). $u(t)$ is the block culture residual of H81: members' content after two-way fixed-effect personal vectors and goal directions are removed. It is an Ornstein–Uhlenbeck vector with fixed variance and rate $k(t)$. A finite magnet has $\alpha_k = -1$; an outside drift has $\alpha_k = 0$. N is the daily active population; time is the active-day index a . R_b is the split-half reliability of block b .

$$du = -k(t) u dt + \sqrt{2k(t)\sigma_u^2} dW, \quad k(t) = k_6 (N(t)/6)^{\alpha_k}$$

$$\tilde{s}_{bb'} = \frac{\langle \cos(u_b^{(4)}, u_{b'}^{(4)}) \rangle}{\sqrt{\hat{R}_b \hat{R}_{b'}}} = A \exp\left(-k_6 \int_{a_b}^{a_{b'}} (N/6)^{\alpha_k} da\right) + s_\infty$$



Observables. (a) Disattenuated similarity of 4-agent culture vectors between blocks of different goals against lag, for pairs inside each village-size class. Only the $N \geq 8$ class lacks the near-lag similarity. (b) Real α_k with delete-one-goal jackknife 95% CIs (dots) against the synthetic drift (D, $\alpha = 0$) and magnet (M, $\alpha = -1$) distributions on the real panel. The two worlds overlap almost completely.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Nothing changes for an operator. One village cannot show whether its culture has inertia that grows with headcount. Regime I holds one slow-mode trajectory of about 13 decay times, and N rises in calendar order. A $1/N$ law in a timescale needs several villages of different size. The useful by-product is post hoc: the slow mode lives in the small-village months (near-lag similarity 0.25–0.28 at $N < 8$, 0.00–0.11 at $N \geq 8$). If it is confirmed, do not expect a village of 10 to carry a month-long culture the way a village of 4–7 did. For H81, split the slow-mode tests at 2025-11-17.

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction	Observed (bge / gte)	Verdict
P1 α_k CI excludes 0, includes -1	$+0.40 [-1.62, 2.41]$ / $+1.22 [-1.45, 3.88]$; power 0.00	inconcl.
Evidence (A3)	LR magnet/drift 0.90 / 0.65; D percentile 0.49 / 0.92	uninformative
P2 NE27 $\Delta \ln k \approx -0.55$	$+0.41$ / $+0.38$ (CIs $\pm 2-4$); power 0.06 / 0.09	inconcl.
P3 full roster more negative	V3 $+0.32$ / $+1.11$ vs V1 $+0.40$ / $+1.22$	weak yes
P4 variants keep sign	V2, V4, V5 flip (synthetic SD 1.2–2.7)	no
N2 #51 joins $\Delta \phi > 0$	NE32 -0.09 / -0.38 ; NE33 $+0.44$ / -0.07 ; power ≤ 0.21	inconcl.
Regime III (descr.)	$+0.33$ / -0.54 , CIs $\pm 7-10$	no info
PH1 era drop (post hoc)	$d_\rho = -0.14$ / -0.28 ; percentile in D 0.11 / 0.00, in M 0.09 / 0.01	gte only

A mapping 1 F identifiability 0
 B assumptions 1 G ground truth 0
 C adequacy 0 H comparative 0
 D unfitted predictions 0 I transfer 0
 E interventional 0
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

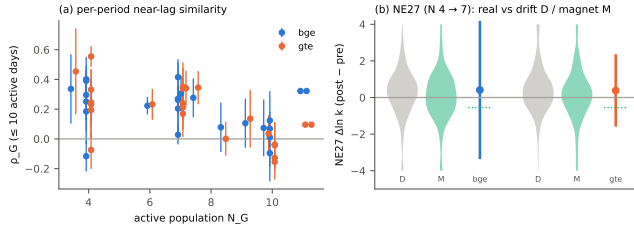
CAVEATS

N is collinear with calendar time: three levels (4, 6–8, 8–12) in date order, with NE03, NE04 and NE08 inside the span. The synthetic assumes one stationary OU mode; the real amplitude changes by era, which neither world contains. The estimator is shifted by $+0.4$ and attenuated to 0.2 of the true slope, though its CI covers the truth in 87–99% of worlds. Per-period ρ_G rests on 1–11 pairs. The era result is post hoc, clear in gte and marginal in bge.

ROUND 2

Confirm the era drop on the held-out goals (#9, #14, #15 vs #22, #28, #29; coordinate with H81). Test inertia cross-sectionally on parallel room trajectories of different size (#38, #44; H108), or across villages.

- **Essence.** whether the slow mode has inertia that scales with the number of carriers (endogenous) or not (outside drift).
- **H106-R1** Era split (feeds H81 round 2).
- **H106-R2** Many trajectories instead of one.
- **H106-R3** Transfer (Q7).



Second figure. (a) Per-period near-lag similarity ρ_G (blocks of other goals within 10 active days) against the active population; bars are ± 1.96 SE. (b) NE27 rate step $\Delta \ln k$ with jackknife CIs against the drift and magnet worlds; dotted ticks mark the magnet value -0.55 .

GOAL PERIODS IN DETAIL

G02 d descriptive
 G03 d descriptive
 G04 d descriptive
 G05 d descriptive
 G06 d descriptive
 G07 d descriptive
 G08 d descriptive
 G10 d descriptive
 G11 d descriptive
 G12 d descriptive

+24 more periods: see the card.